SIMON FRASER UNIVERSITY
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School of Engineering Science

ENSC 325
Final Project
Bandgap Reference Voltage

Group	8
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Abstract

This project aims to develop a functional bandgap reference voltage capable of withstanding changes in temperature. Primary objectives include researching techniques to achieve our goal, implementing calculations to meet our requirements, and producing simulations in Cadence OrCAD that function as desired. The bandgap reference voltage performance will be assessed based on the ability to generate a stable reference voltage of about 1.2V, minimizing the temperature coefficient and low power consumption. The results provide insight into the bandgap reference voltage, how it's created, implemented, and optimised.

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1.0 Initial Research

1.1 Objective and Constraints

The primary objective of this project is to design and simulate a CMOS-based Bandgap Reference (BGR) circuit that delivers a stable output voltage, independent of temperature and supply voltage variations. The design should ensure minimal temperature drift, low power consumption, and reliable operation under varied environmental and supply conditions. The project aims to deepen the understanding of analog circuit design and simulation using OrCAD X software.

A central goal is to achieve a highly stable output voltage with a temperature coefficient of less than or equal to 100 ppm/°C, ensuring consistent performance across a wide temperature range. The design should integrate effective temperature compensation techniques, such as Proportional to Absolute Temperature (PTAT) and Complementary to Absolute Temperature (CTAT) current strategies, to balance thermal effects. Additionally, the circuit must be power-efficient, operating with minimal current draw to support use in energy-sensitive environments, such as in IoT applications. Throughout the project, OrCAD X tools are used to simulate the circuit's behavior under various conditions and refine the design to optimize power efficiency, accuracy, and robustness.

Several constraints must be considered throughout the design process. First, the circuit must be built using the provided 5µm BNR CMOS transistor models, adhering to the specific electrical parameters defined for both NMOS and PMOS devices. All simulation and analysis must be conducted within the OrCAD X platform, which serves as the standard environment for validating circuit performance. Design decisions must address power consumption limits, ensuring the final circuit operates efficiently without compromising accuracy or thermal stability. Furthermore, the BGR circuit must consider reliable biasing and startup circuitry to guarantee stable operation from initial power-on.

1.2 Research

Multiple scholarly sources were consulted, providing several viable options for the topology of the BGR circuit. For the sake of design simplicity, options were narrowed down to Brokaw cell and current mirror designs. Both topologies have been extensively used in literature and practical applications for generating temperature-independent reference voltages. The Brokaw cell is a well-established configuration known for its precise voltage output and good temperature compensation, typically combining bipolar junction transistors (BJTs) and operational amplifiers to produce a stable reference. However, this topology generally involves a more complex feedback and biasing network, requiring precise channel width and length matching of devices and greater design effort to ensure startup reliability and robustness.

On the other hand, current mirror-based BGR architectures offer a more streamlined approach, particularly within CMOS design environments. These designs utilize temperature-dependent current generation (PTAT and CTAT) and exploit current mirrors to sum or balance these currents, achieving the desired reference voltage with reduced component count and simpler layout considerations. Current mirror topologies also integrate more naturally into low-power, compact systems, which aligns with the project’s secondary goal of minimizing power consumption.

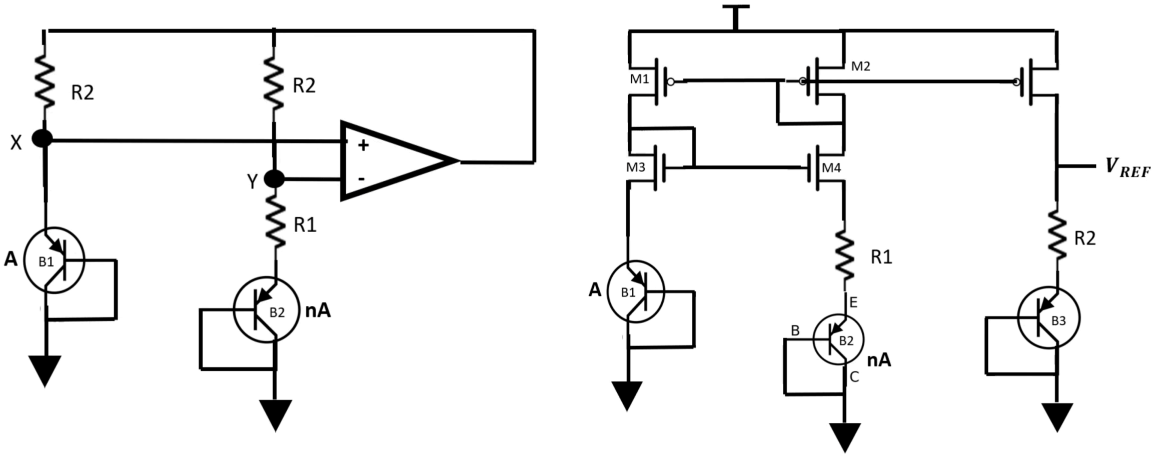


Figure 1: Brokaw cell (left) and Widlar current mirror (right) topologies.

Source: Adapted from [1]

After weighing the trade-offs, the current mirror design was selected due to its relative simplicity, ease of implementation in CMOS technology, and suitability for simulation in the OrCAD X environment. While it may not match the theoretical precision of a Brokaw cell under all conditions, its simplicity allows for more efficient design iterations, easier debugging, and sufficient accuracy for the project's targeted performance metrics. This decision reflects a balance between academic rigor and practical design constraints, ensuring that the final design remains both feasible and effective within the scope of the project timeline and deliverables.

2.0 Theory & Calculations

2.1 Bandgap Core

The primary goal of the Bandgap Core is to develop a circuit that is capable of generating a constant Vref with a variable temperature. To do this, we will make use of the PTAT (proportional to absolute temperature) and CTAT (complementary to absolute temperature) characteristics of the PN junction on the diode-connected BJT.

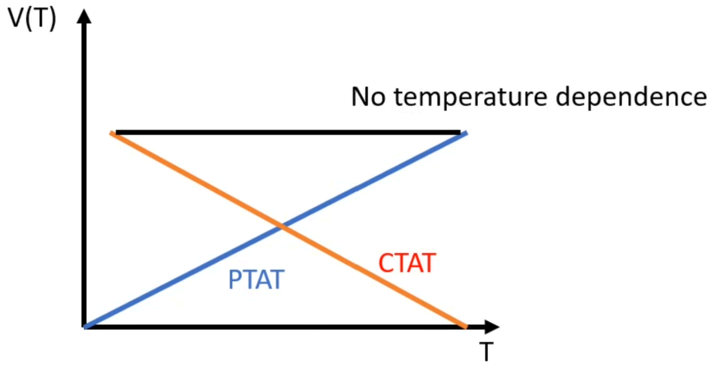


Figure 2: PTAT, CTAT, and combined V(T) over temperature.
Source: Adapted from [1]

The development of CTAT and PTAT components requires a means of gauging temperature through the use of diode-connected BJTs. It is known that the base-emitter voltage of a BJT, v_{BE} is proportional to the thermal voltage, V_T, which is related to temperature through the following equations:

$$v_{BE} \approx V_T \ln \frac{I_C}{I_S} \qquad V_T = \frac{KT}{q}$$

Where T is temperature in Kelvin, I_c is collector current, I_s is scale current, K is the Boltzmann constant, and q is the elementary charge. To find the variation of V_{be} as a function of temperature, the derivative of V_{be} with respect to T is evaluated. Assuming a linear relationship between I_c and T, the following equation is derived:

$$\frac{D\,v_{BE}}{DT} = \frac{v_{BE} - \frac{3}{2}V_T - \frac{E_g}{q}}{T}$$

Where E_g is the bandgap energy of silicon. This derivative is complementary to absolute temperature. A practical implementation of the circuit shown in Fig. 3 can be used to derive a parameter that is proportional to absolute temperature.

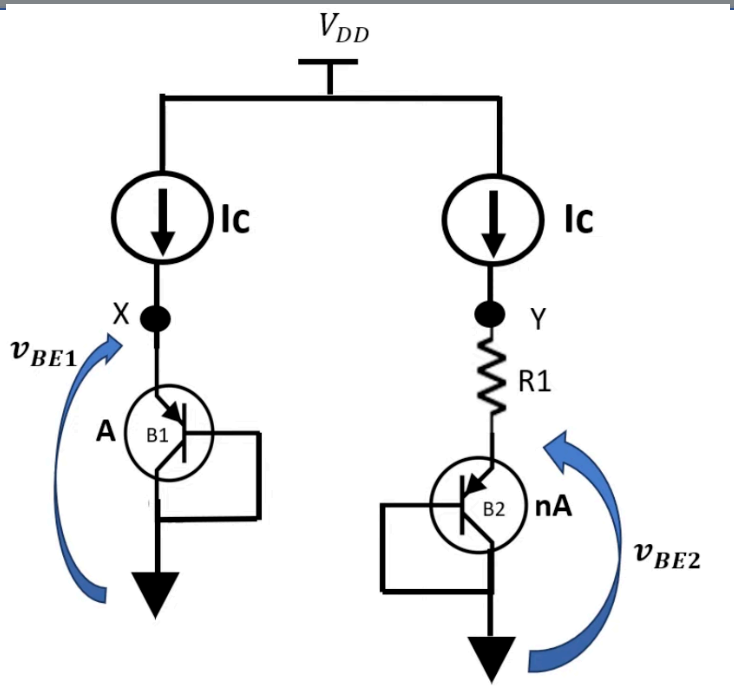


Figure 3: Bandgap core ideal circuit.
Source: Adapted from [1]

The difference in base-emitter voltages between transistors B1 and B2 can be found using the formula $\Delta v_{be} = V_T \ln(n)$, where n is the area scale factor of B2. Taking the derivative of this term with respect to temperature provides a result that is proportional to absolute temperature:

$$\frac{D\Delta v_{BE}}{DT} = \frac{K}{q} \ln n$$

The following equations are derived to find the voltage at node Y in the circuit observed in Fig. 3 :

$$V_X = V_{BE1} \qquad V_Y = V_{BE2} + \text{R1 Ic}$$

$$V_{BE1} = V_{BE2} + \text{R1 Ic}$$

$$V_{BE1} - V_{BE2} = \text{R1 Ic} = V_T \ln n$$

$$V_Y = V_{BE2} + V_T \ln n$$

Ideally, the voltage at node Y is to remain constant and be independent of temperature drift. Therefore the derivative of V_Y with respect to temperature must be zero. To achieve this, the scale factor n can be adjusted. Although this approach would result in an impractical scale factor in the order of 10^7. An alternative, more practical approach sees the introduction of an additional third output branch as seen in Fig. 4. This branch contains resistor R2 and a third transistor with an area equal to that of B1.

A practical circuit can now be developed. To create the current sources, two PMOS transistors, M1 and M2, and two NMOS transistors, M3 and M4 are used. The use of a cascode arrangement mitigates the effect of mismatched drain voltages between M1 and M2 and increasing output impedance, ensuring a relatively stable current output. The center branch provides PTAT bias current $V_T ln(n) / R1$.The output branch, connected through mirroring PMOS M8, mirrors this current through R2 and B3. A stable reference voltage can then be observed at the junction between the drain of M5 and R2 following the expression seen in Fig. 4.

$$V_{REF} = V_{BE3} + \frac{R2}{R1}\ V_T \ln n$$

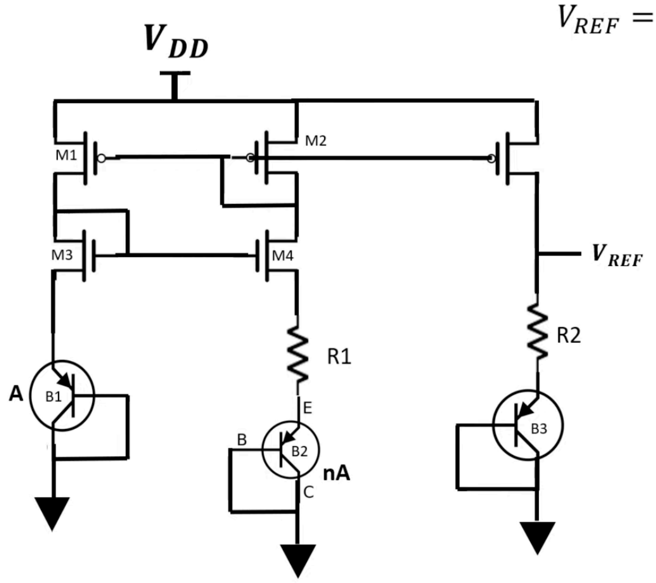


Figure 4: Practical BGR circuit implementation using current mirrors
Source: Adapted from [1]

2.2 Startup Circuit

The functionality of the circuit observed in Fig. 4 is dependent on the flow of current in all 3 branches, though problems arise during startup when power is initially switched on. Upon switching the sources and gates of M1, M2 and M8 are pulled up to V_{DD}, preventing the flow of current and leaving the rest of the circuit pulled down to ground. The implementation of a startup circuit seen in Fig. 5 ensures proper operation after power is switched on.

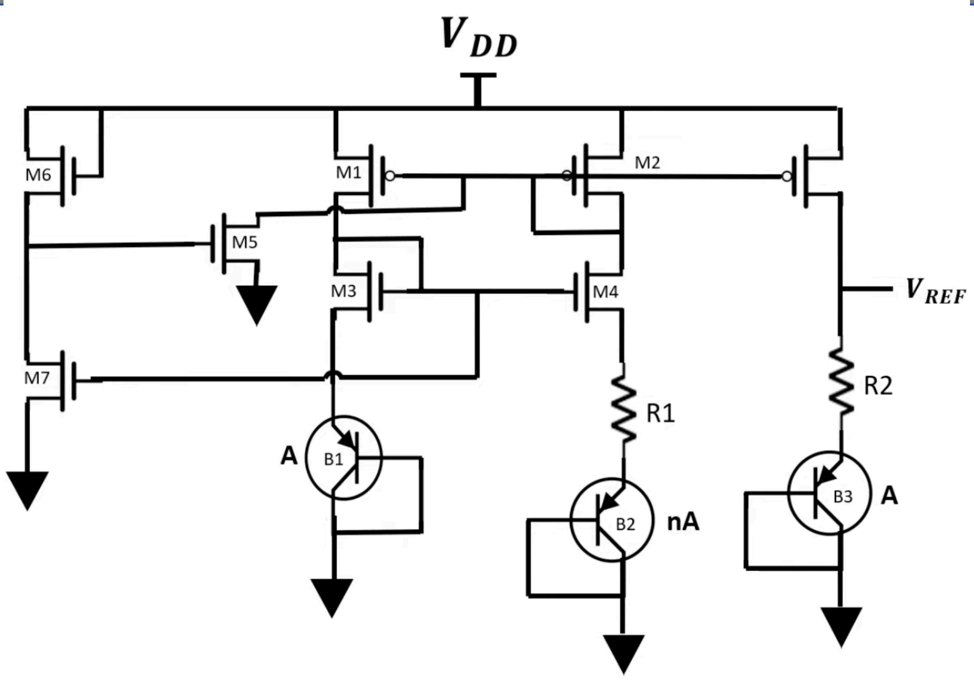


Figure 5: Bandgap reference with Start-Up circuit
Source: Adapted from [1]

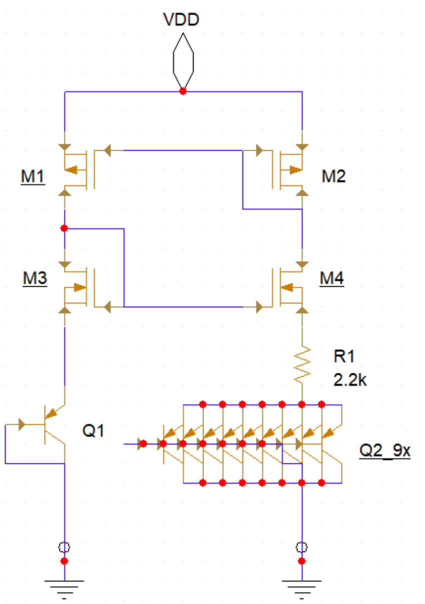
PMOS transistors M5 and M6 ensure the gate voltages of M1, M2 and M8 are pulled down on startup. Initially, M7 is off due to the gate voltages of M3 and M4 being pulled down to ground. This provides a voltage of V_{DD} at the gate of M5, turning it on and pulling down the gate voltages of M1, M2 and M8. Once these transistors turn on, the gates of M3, M4, and M7 are pulled up. With M7 turned on and current flowing through the leftmost branch, the gate voltage of M5 drops significantly, turning it off and effectively disconnecting the startup circuit from the current mirror. The drop in voltage at the junction between M6 and M7 can be emphasized by increasing the channel area of M7, making it more conductive relative to M6.

3.0 Simulation

3.1 OrCAD X Circuit

SPICE models for the 5μm CMOS transistors used in the current mirrors and start-up circuit were provided in the lab documentation. Additionally, SPICE models for the 2N3906 PNP BJTs used in the CTAT and PTAT portions of the circuit were sourced from the DigiKey website [2].

Circuit construction began with the addition of the cascoding Widlar current mirror using one pair of NMOS transistors, M1 and M2, and another pair of PMOS transistors, M3 and M4, as seen in Fig. 5. The output of the left side of the current mirror was connected through a single diode-connected PNP BJT, Q1, to ground. The output of the right side was connected through biasing resistor R1 in series with a set of 9 parallel diode-connected PNP BJTs, Q2, to ground. This completed the PTAT current biasing branch.



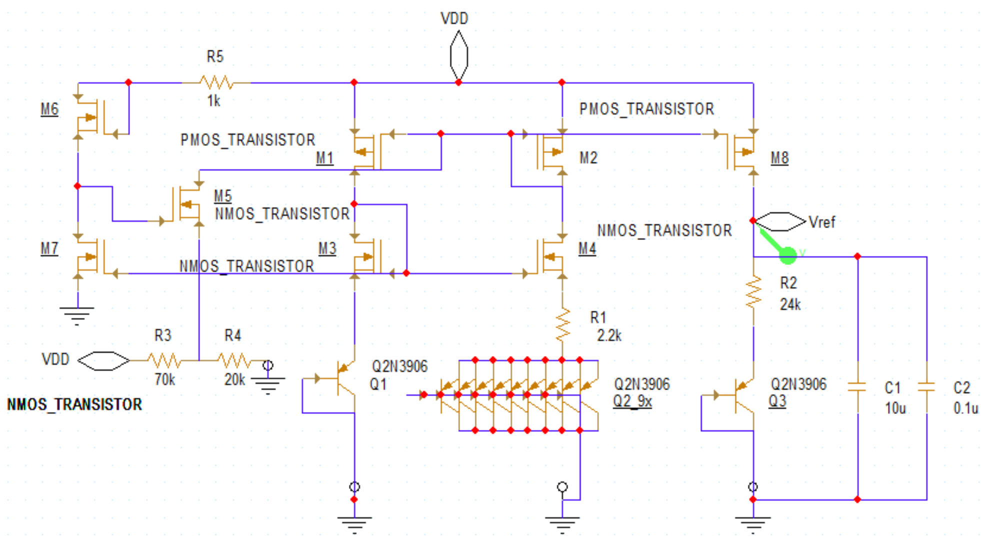


Figure 5: Current mirror and PTAT biasing circuit

Figure 6: Finalized BGR circuit

A third output branch was added to the right with a PMOS transistor, M8, to mirror the PTAT bias current through resistor R2 in series with a final diode-connected PNP BJT, Q3, to ground, thus creating a reference voltage node between M8 and R1. Lastly, startup circuitry was added as illustrated in Fig. 6. NMOS M7 was created using 6 transistors stacked in series to increase channel area by a factor of 6, effectively reducing resistance.

Initial values for resistors R1 and R2 as well as the scale factor n for Q2 were roughly estimated by use of the expressions determined in the development of the bandgap core, and adjustments were made as testing was conducted.

3.2 Temperature Sweep Test

The priority in the first stages of testing was to minimize the effects of temperature on the output reference voltage. Initial testing provided promising results seen in Fig.

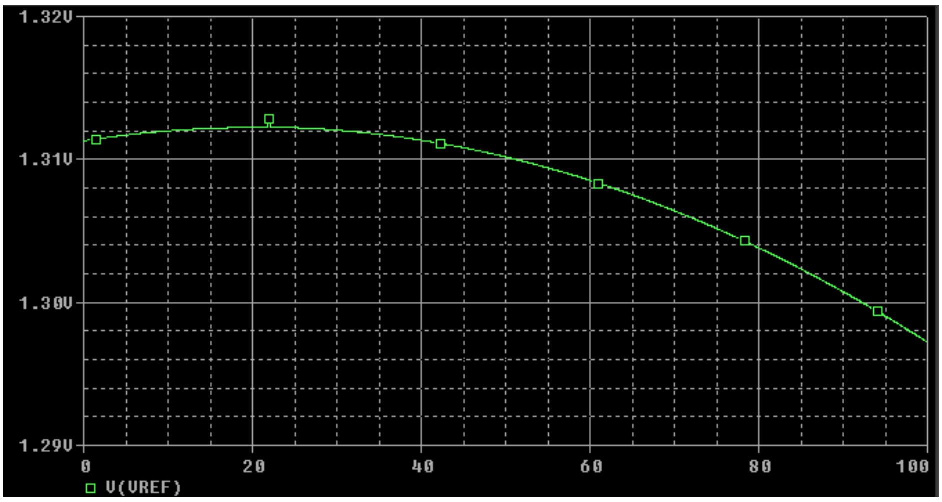


Figure 7: V_{Ref} vs Temperature from 0 to 100°C

To quantify the temperature dependence of VREF, the temperature coefficient (TC) was calculated using the following formula:

$$TC_{ppm} = \left|\frac{VRef(T2)-VRef(T1)}{VRef-nom \times (T2-T1)} \times 10^{6}\right| = \left|\frac{1.298-1.311}{0.5 \times (1.298+1.311) \times (100-0)} \times 10^{6}\right| = 99.6\, ppm/C$$

This result indicates that the circuit exhibits good temperature stability, with a TC well below 100 ppm/°C, which is typical for well-designed bandgap references.

3.3 Line Noise Test

In this test, a 1 VAC signal was superimposed onto the DC supply rail to emulate real-world conditions such as electromagnetic interference (EMI), electrochemical coupling (ECC), or switching noise commonly present in SMPS-powered systems. This scenario demonstrates how the bandgap reference circuit performs under supply ripple disturbances.

To mitigate the resulting ripple at the output, two decoupling capacitors — 10 µF and 0.1 µF — were connected from VREF to ground. The combination of these capacitors filters out both low-frequency and high-frequency noise components.

As shown in Figure 8, the VREF signal before decoupling (left) exhibits significant ripple due to the injected AC noise. After installing the capacitors (right), the ripple is almost entirely suppressed, resulting in a clean and stable DC output. This confirms the effectiveness of decoupling capacitors in enhancing power supply rejection.

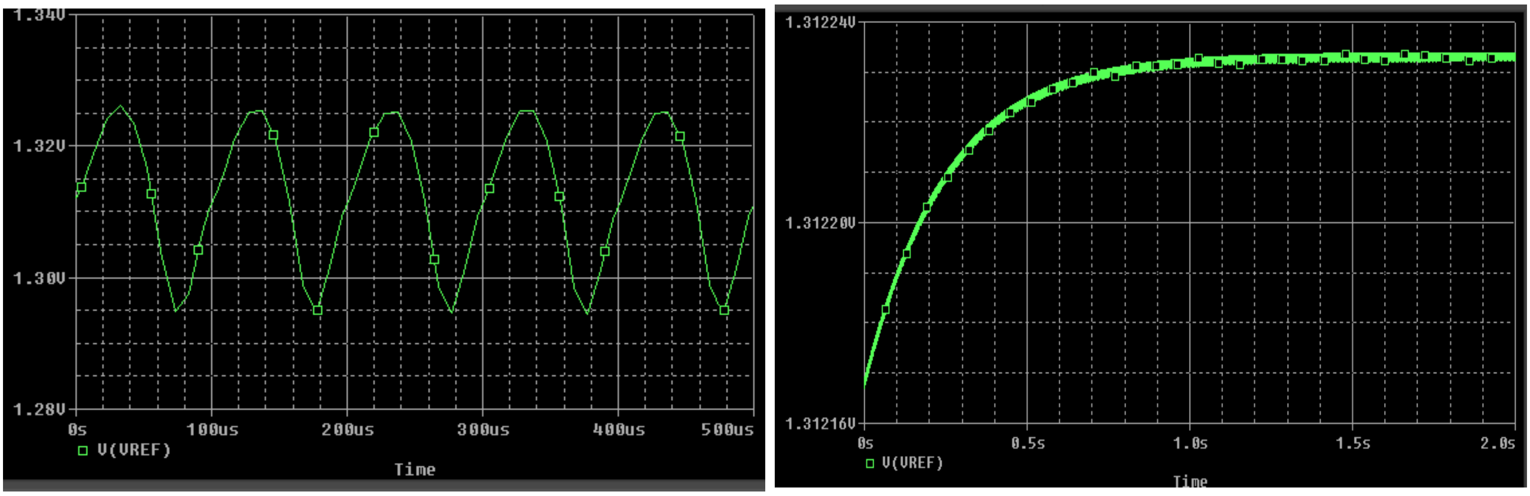


Figure 8: V_{Ref} before (Left) and after (Right) installing the decoupling capacitors C1, C2

3.4 Power Consumption

To evaluate the efficiency of the bandgap reference circuit, the power consumption was measured at only 5mW by monitoring the DC supply current. This power level is considered low and acceptable for IoT applications, where energy efficiency is crucial. The design achieves a good balance between maintaining circuit stability and minimizing power consumption. Reducing resistor values or using lower supply voltages could further improve efficiency, depending on application requirements.

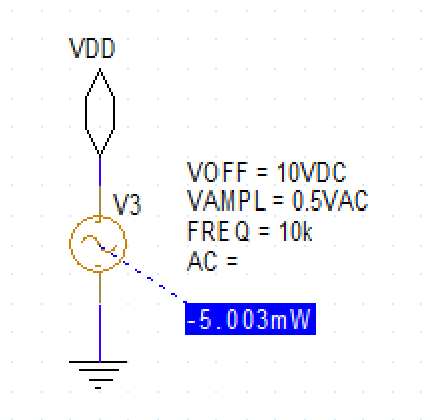


Figure 8: Power Consumption of the circuit

4.0 Conclusion

This project successfully implemented and tested a bandgap reference voltage circuit using a CMOS-compatible current mirror architecture and startup enhancement. The circuit was designed to produce a temperature-insensitive reference voltage.

Several simulations were conducted to validate performance. Temperature Sweep testing demonstrated a temperature coefficient of 99.6 ppm/°C, well within the 100 ppm/°C target range. Line Noise testing verified the effectiveness of decoupling capacitors in suppressing supply ripple, resulting in a clean VREF even under AC injection. Power Consumption analysis confirmed that the design consumes just 5 mW, suitable for low-power applications like IoT.

Through iterative tuning of resistor values and proper application of PTAT and CTAT principles, the circuit achieves a good trade-off between stability, accuracy, and efficiency. Future improvements could include adding curvature correction or further optimizing startup behavior at low temperatures.

Overall, the project deepened the team's understanding of analog design, temperature compensation, and CMOS simulation workflows in OrCAD X.

References

[1] Computer&Electronics, *bandgap reference - voltage reference - voltage source - start-up circuit*, (Nov. 04, 2024). Accessed: Apr. 10, 2025. [Online Video]. Available: https://www.youtube.com/watch?v=UUoxYuPRXhA

[2] "2N3906 PBFREE," DigiKey Electronics. Accessed: Apr. 10, 2025. [Online]. Available: https://www.digikey.ca/en/products/detail/central-semiconductor-corp/2N3906-PBFREE/4806878